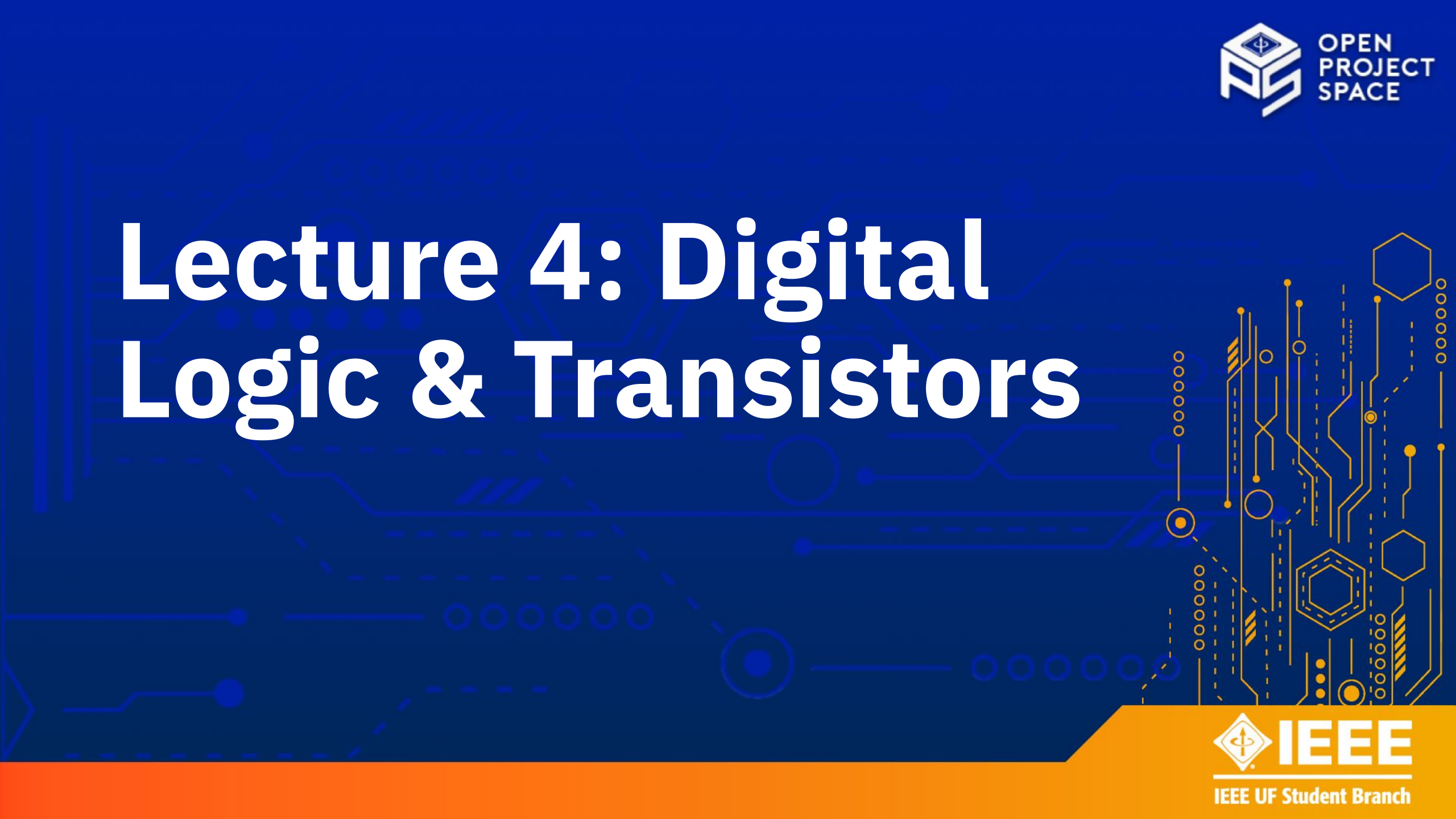
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Lecture 4: Digital Logic & Transistors



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Overview

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- Introduction to basic Digital Logic
- What are transistors?
- How do transistors work?
- How can we use transistors in our circuits?
- Project 4: SR Latch

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Basic Logic

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Why do we Need Logic?

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- Forms the backbone of computers, processors, and embedded systems
- Bridges software and hardware understanding
- Powers real-world technology

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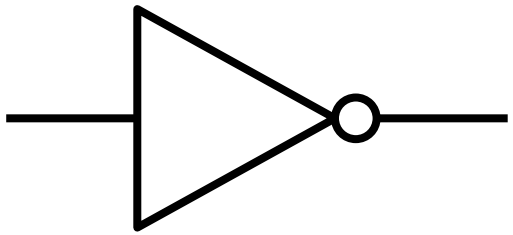
Boolean Logic

Input 1	Input 2	Output
False/F (0)	False/F (0)	?
False/F (0)	True/T (1)	?
True/T (1)	False/F (0)	?
True/T (1)	True/T (1)	?

- There are only two states: True/1 or False/0
- Truth tables show the output of a logic circuit for every possible input

NOT (Inverter)

Input	Output
F	T
T	F



- Negates or Inverts a value
- NOT uses ‘!’ to negate a value in C++/Arduino
- Example: false != true

OR



Input 1	Input 2	Output
F	F	F
F	T	T
T	F	T
T	T	T

- Is true if EITHER input 1 OR input 2 is true
- It does not matter which of the two are true, so long as one of them is
- OR uses ‘||’ in C++/Arduino code
- Example: input1 || input2

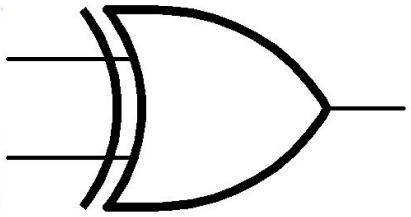
AND



Input 1	Input 2	Output
F	F	F
F	T	F
T	F	F
T	T	T

- Is true if BOTH input 1 and input 2 are true
- Both inputs MUST be true
- AND uses && in C++/Arduino code
- Example: input1 && input2

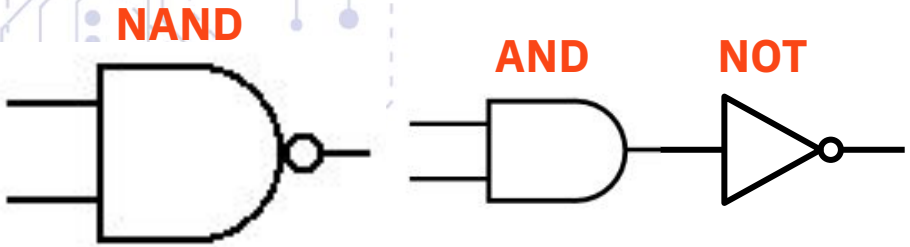
XOR



Input 1	Input 2	Output
F	F	F
F	T	T
T	F	T
T	T	F

- Is true if inputs are NOT EQUAL
- Both inputs MUST be opposites
- XOR uses != in C++/Arduino code
- Example: input1 != input2

Using NOTs with Other Gates



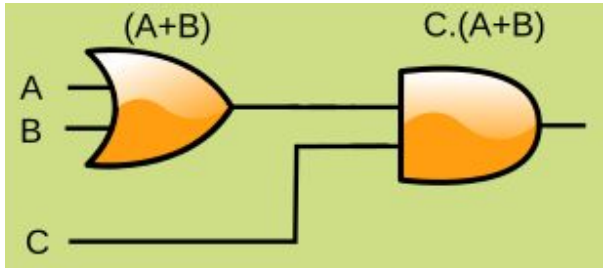
Input 1	Input 2	AND	NAND
F	F	F	T
F	T	F	T
T	F	F	T
T	T	T	F

- **NOT gates can negate the output of any logic gate**
- **These negations have unique names**
- **For example:**
  - AND -> NOT = **NAND**
  - NOR -> NOT = **NOR**
  - XOR -> NOT = **XNOR**

Logic Equations

C	B	A	C*(A+B)
F	F	F	F
F	F	T	F
F	T	F	F
F	T	T	F
T	F	F	F
T	F	T	T
T	T	F	T
T	T	T	T

- Logic gates can be connected together to make larger circuits & components
- As a result, we get a longer logic equation
- The truth table may get tall...

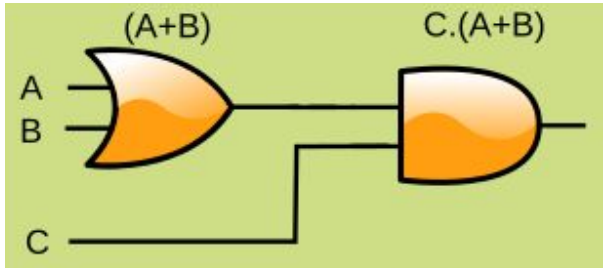


Logic Equations (simplified)

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C	B	A	A+B	C*(A+B)
F	*	*	*	F
T	F	F	F	F
T	*	T	T	T
T	T	*	T	T

- Can identify intermediate outputs to make solving logic equations easier
- Sometimes inputs can be ignored (don't cares)
- These things can make creating truth tables much easier!



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Logic Identification

Input 1	Input 2	Output
F	F	T
F	T	F
T	F	F
T	T	F

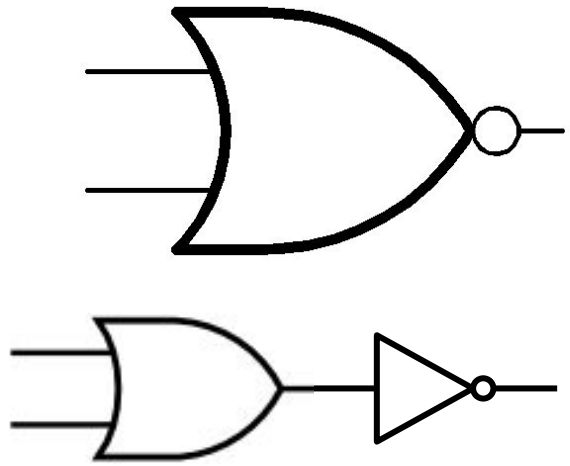
What is the logic gate or equation that is correctly associated with this truth table?

Logic Identification

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It’s a NOR gate or OR -> NOT

Input 1	Input 2	OR	NOR
F	F	F	T
F	T	T	F
T	F	T	F
T	T	T	F



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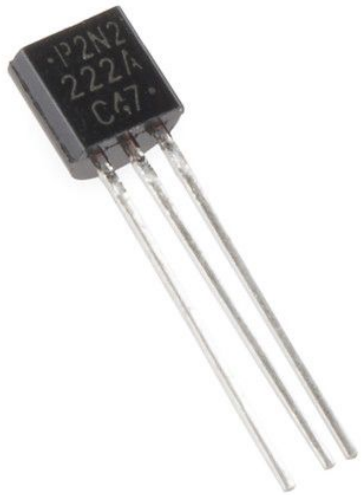
Transistors

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Preface

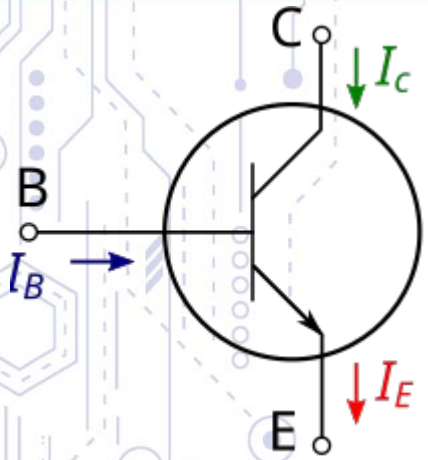
- Transistors, and semiconductors as a whole, are very complicated
- This information is meant to give you a broad overview of transistors, what they do, and how they can be applied in real circuits
- You may not fully understand everything immediately, and that’s okay and expected! There are resources on the Github, office hours, and other OPlings to help!
- The information in this lecture will be enough for you to understand how to do the project

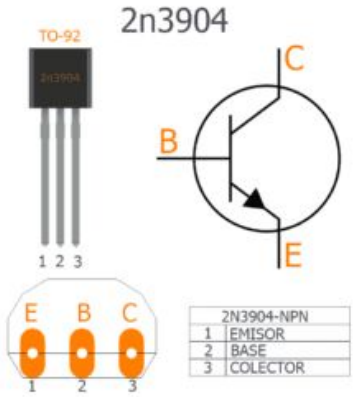
Transistors



- Transistors are electronic switches
- They have two current terminals and one control terminal
- Based on the current or voltage on the control terminal, the other two terminals are electrically connected

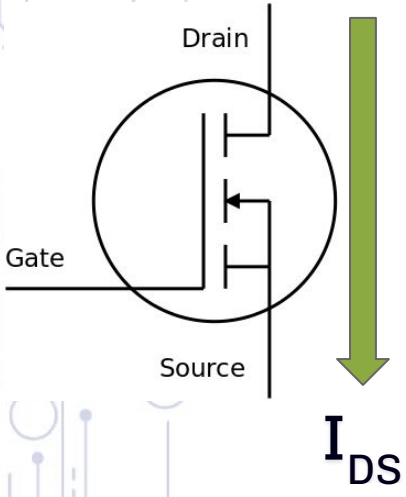
Transistor Types: BJT



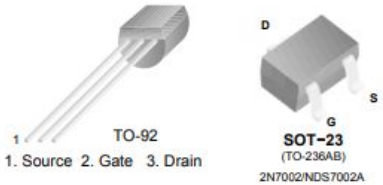


- **Bipolar Junction Transistors (BJTs) are controlled by a current on their Base terminal**
- **When a sufficient current flows from the Base to the Emitter, current is able to flow from the Collector to the Emitter**

Transistor Types: FET



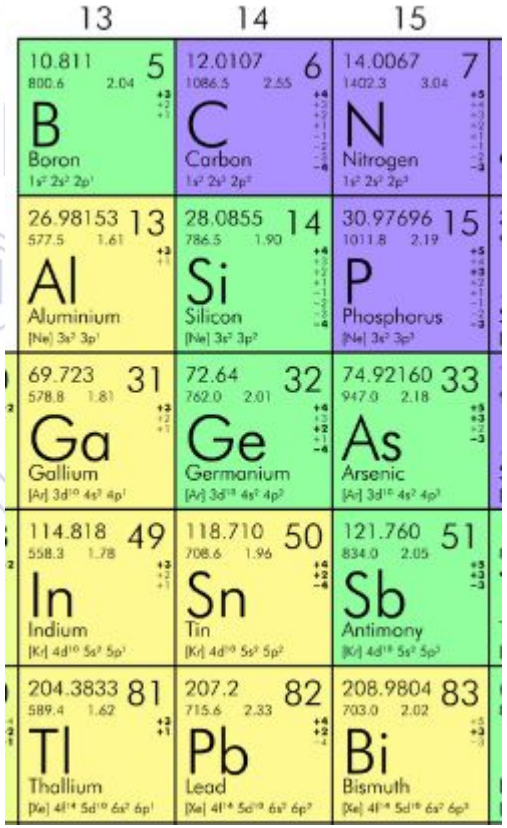
2N7000 / 2N7002 / NDS7002A



- Field Effect Transistors (FETs) are controlled by a voltage on their Gate terminal
- When a sufficient voltage is applied to the Gate, current is allowed to flow from Drain to Source
- We will be focusing on MOSFETs (Metal Oxide Semiconductor Field Effect Transistors) in this lecture

How MOSFETs function

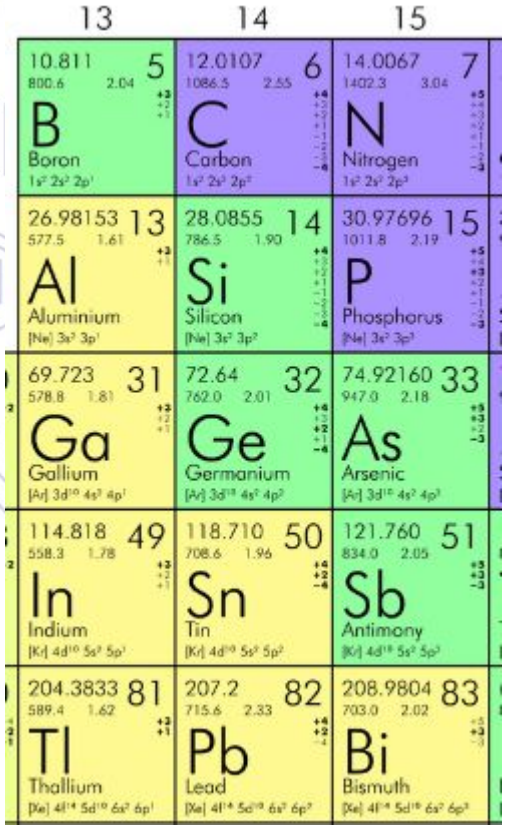
Semiconductor Doping

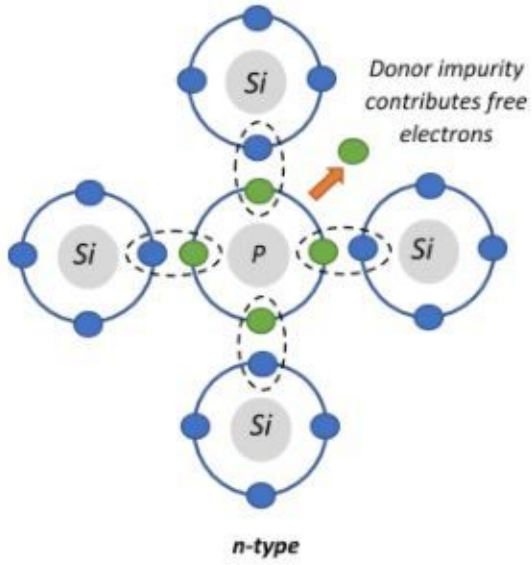


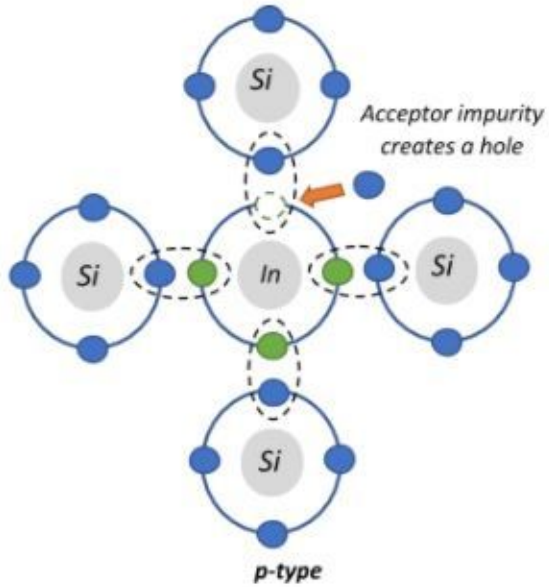
- Semiconductors can act as conductors or insulators depending on their structure
- Semiconductors are fabricated on a silicon wafer
- Impurities can be added to the silicon to give the silicon a set of desired electrical properties

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Semiconductor Doping

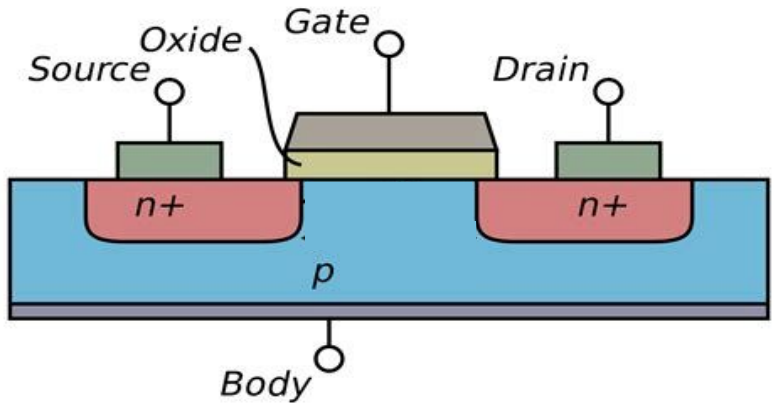






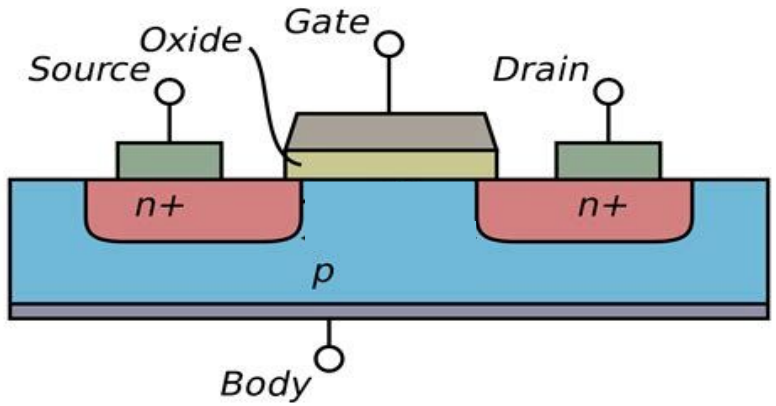
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N-MOSFET Architecture



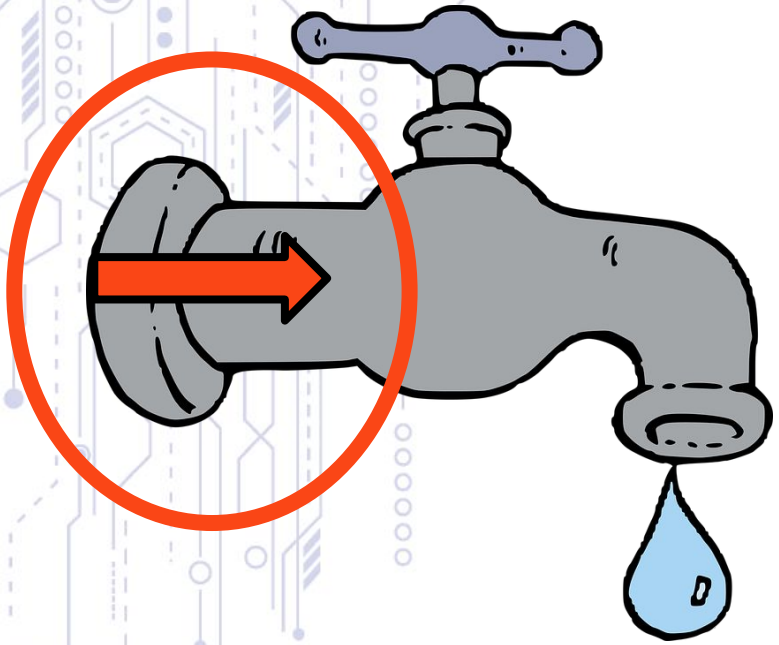
- This lecture, we will be looking at N-MOSFETs, which use electrons as charge carriers
- Their drain and source terminals are heavily doped n-silicon, with p-type silicon in between

MOSFET Mechanics



- There are two important quantities that determine how an NMOS behaves: the Drain-Source Voltage (V_{DS}) and the Gate-Source Voltage (V_{GS})

Drain-Source Voltage



- Drain-Source voltage is like the water pressure behind a spigot
- Higher V_{DS} means more current FLOWS through the MOSFET
- However, although the water pressure might be high, the spigot must be opened enough to let the water flow through

Gate-Source Voltage



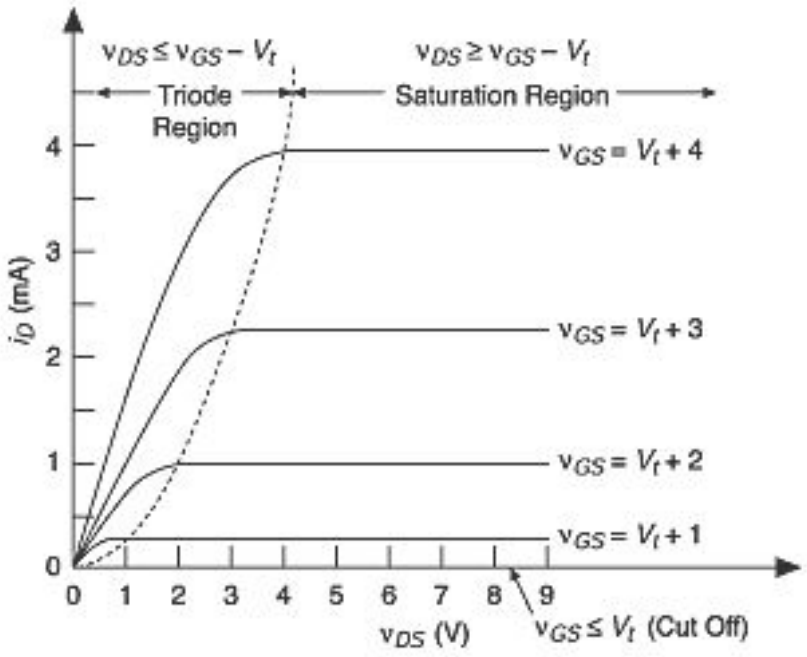
- Gate-Source voltage is like the knob on the spigot
- Higher V_{GS} means more current is ALLOWED to flow through the MOSFET
- However, there must be sufficient water pressure to push the water through at a fast flow rate

Water Analogy Summary



- Drain-Source voltage is like the water pressure behind a spigot
- Gate-Source voltage is like the knob on the spigot
- Higher V_{DS} means more current FLOWS through the MOSFET
- Higher V_{GS} means more current is ALLOWED to flow through the MOSFET
- These two voltages constrain the transistor's current flow

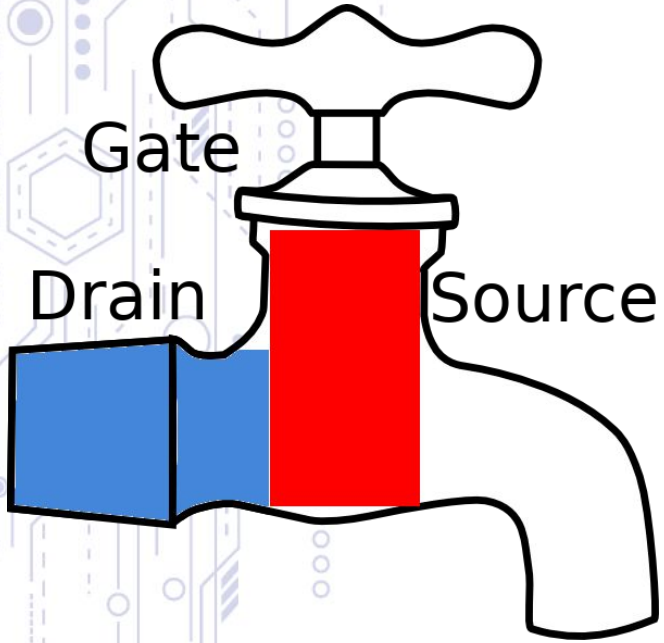
Transistor Regions



- MOSFET behavior is categorized into three regions based on V_{GS} and V_{DS}
- These regions are:
  - Cutoff
  - Linear
  - Saturation

Cutoff Region

$V_{GS} < V_T$

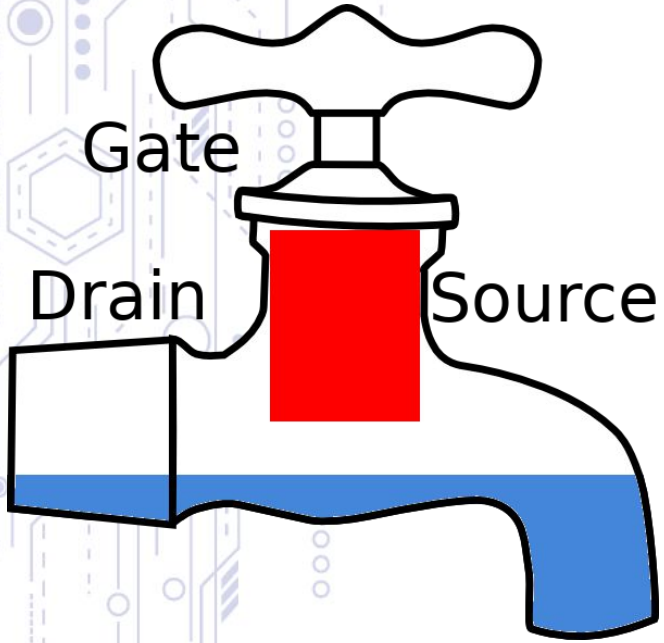


- Gate-Source voltage is too low to allow current to flow from drain to source
- For the current to be able to flow, V_{GS} must be greater than the threshold voltage (V_T)
- The threshold voltage is the characteristic of the transistor that controls whether the transistor is on or off

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Linear Region

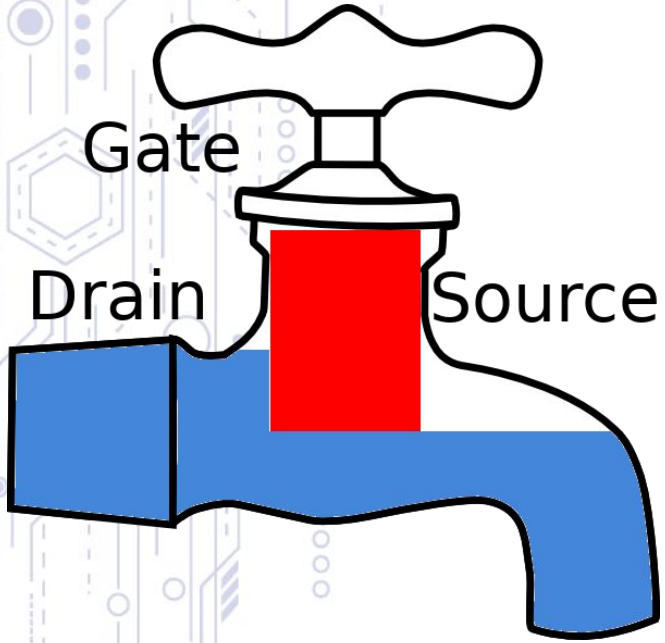
$V_{GS} \geq V_T$ & $V_{GD} \geq$ V



- Gate-Source voltage is above the threshold voltage ($V_{GS} > V_T$)
- I_{DS} is proportional to V_{DS}
- Spigot openness is NOT constraining flow

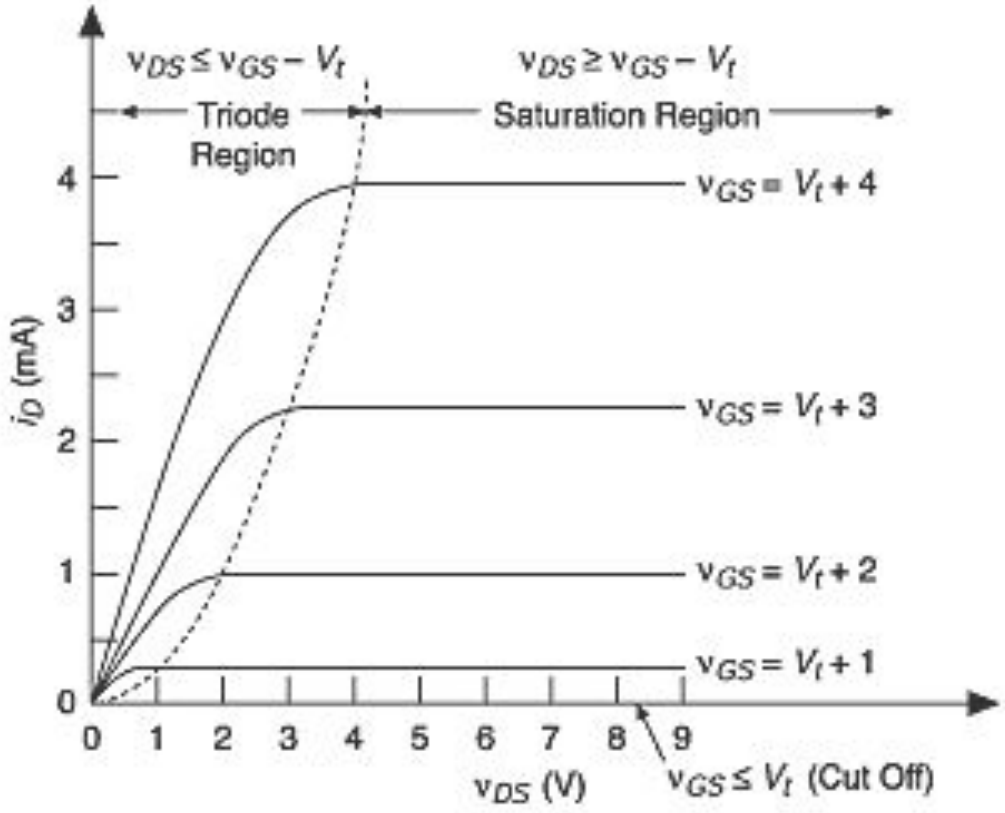
Saturation Region

$V_{GS} \geq V_T$ & $V_{GD} <$ V



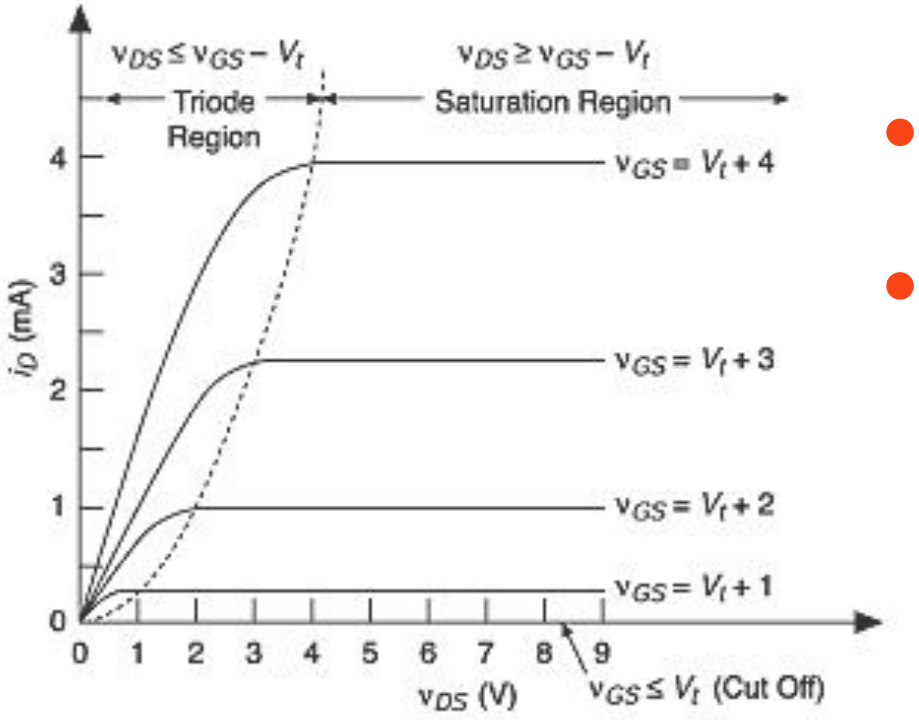
- Drain-Source current is at its maximum for the current V_{GS} and changes in V_{DS} do not affect it
- If you opened the spigot more, you could get more current through

Transistor Regions



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Transistor Regions Summary



- Cutoff: No current flow, $V_{GS} < V_T$
- Linear: Constrained by V_{DS}, $V_{GS} \geq V_T$ & $V_{GD} \geq V_T$
- Saturation: Constrained by V_{GS}, $V_{GS} \geq V_T$ & $V_{GD} < V_T$

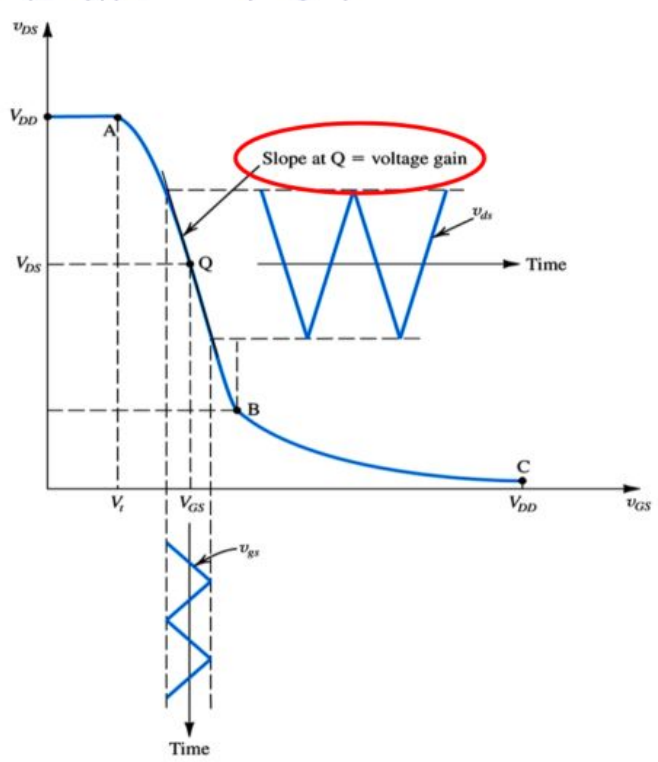
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Transistor Applications

Amplification

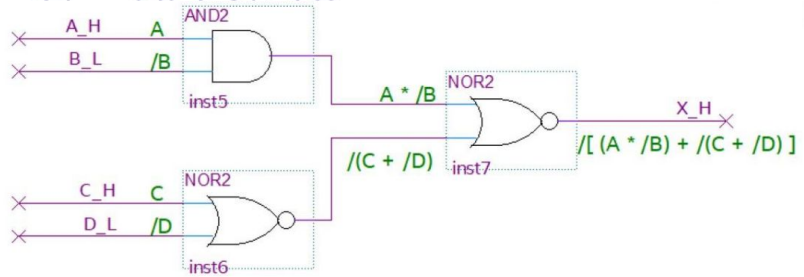
- Amplifiers are electronic devices that increase the voltage, current, or power of a signal
- This makes it easier for us to measure very weak signals
- It can also keep signals strong when transmitting over a long distance, such as for internet or radio

Amplification



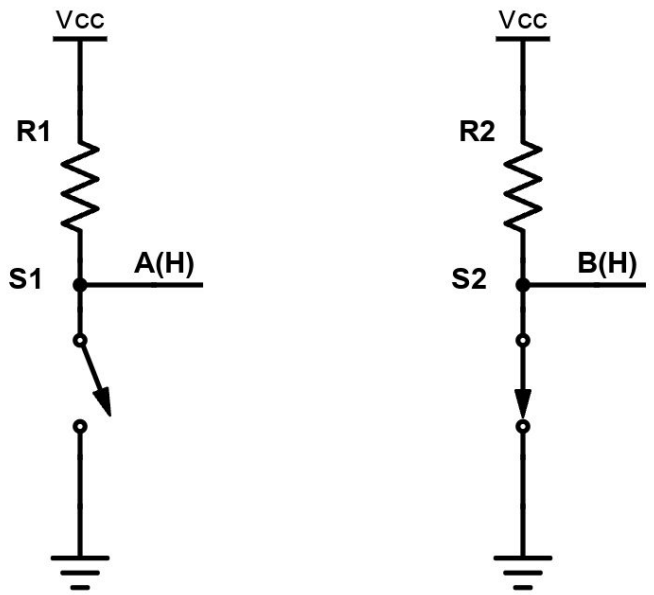
- Amplification uses the saturation region of a transistor
- A small variation in the gate voltage in the saturation region leads to a large change in the drain-source current

Digital Logic Circuits



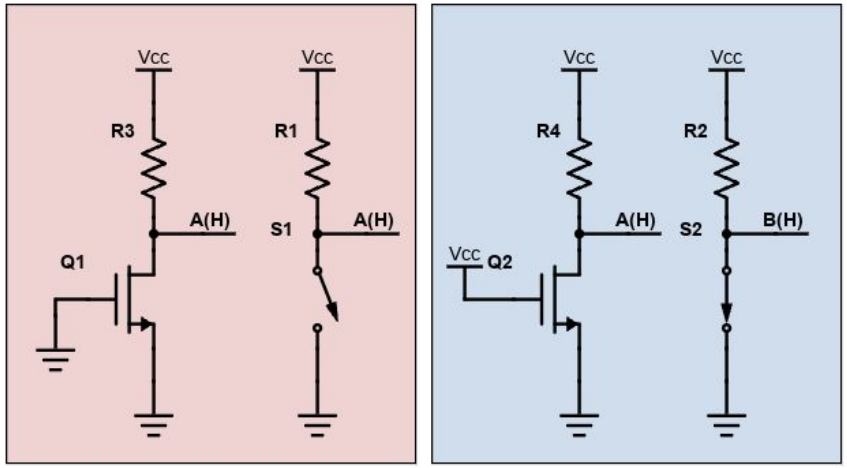
- Using transistors as switches allows us to control circuits and create the logic gates that we discussed earlier!
- Underneath the nice-looking gate symbols, transistors are what allow the gates to function as they do
- Transistors form digital logic, which forms the computing systems we have today!

Switch Circuit Refresher



- When the switch is open, no current flows, and output A remains at VCC
- When the switch is closed, output B is directly tied to ground

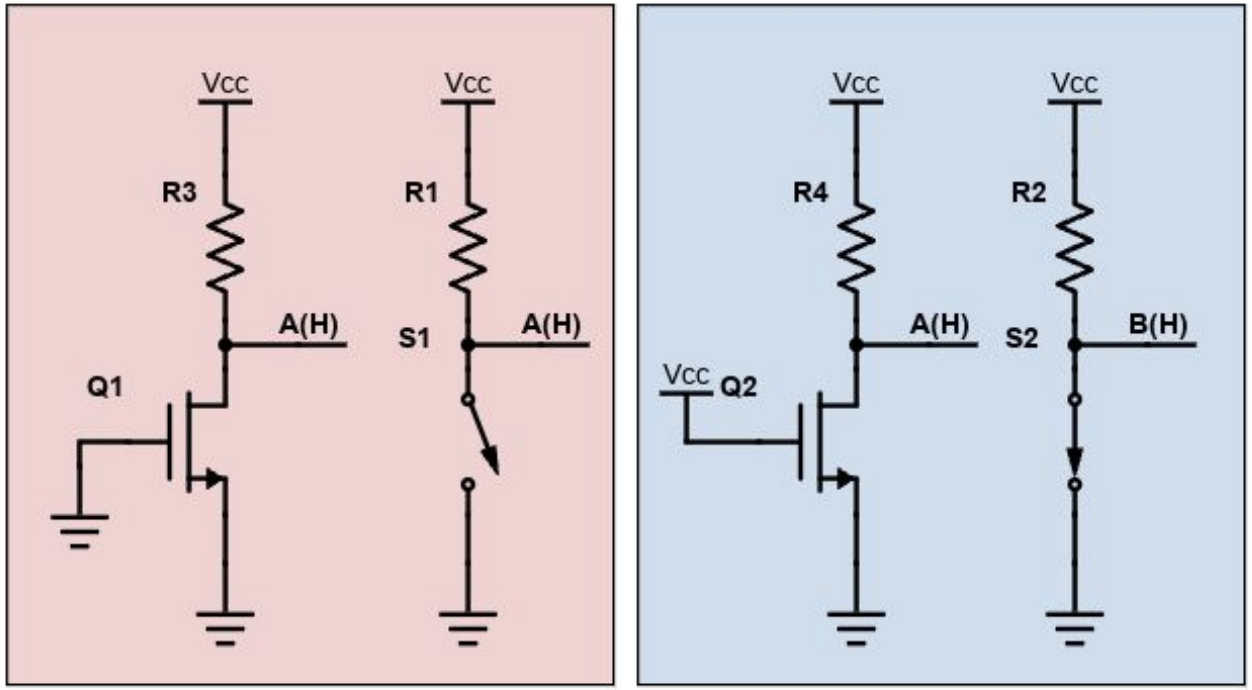
NMOS as Switch



- We can make the same circuit using a MOSFET!
- No current flows when $V_{GS} < V_T$ (cutoff)
- When a high V_{GS} is applied, current is allowed to flow (linear)

NMOS as Switch

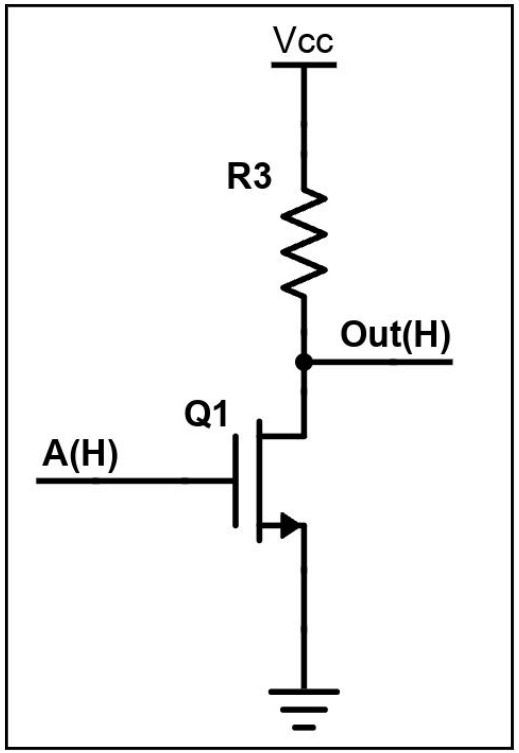
$V_{GS} < V_T$, no current flow, cutoff region



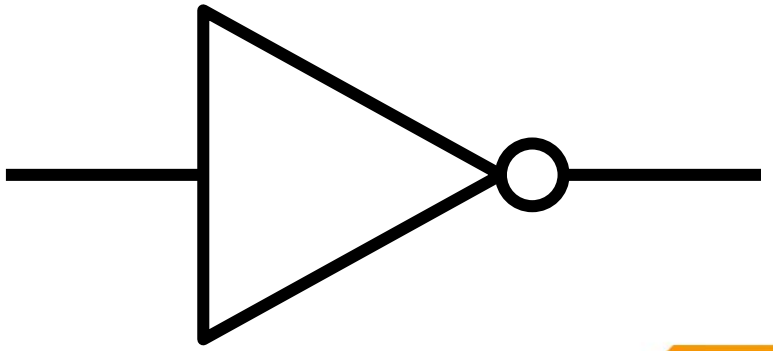
V_{GS} high, current can flow, linear region

Basic Transistor Digital Logic

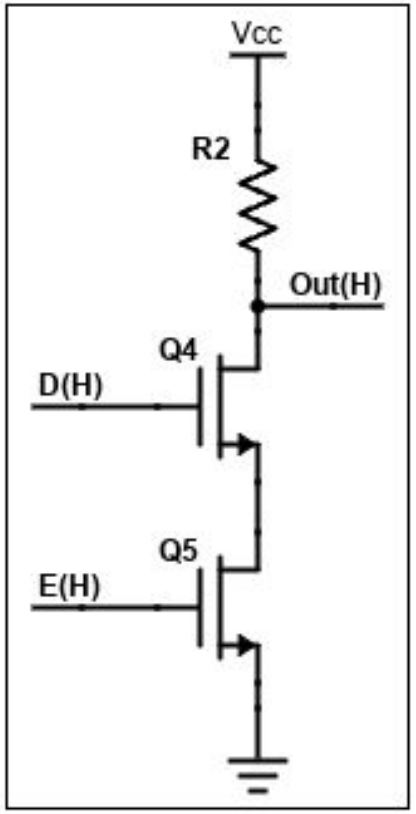
NMOS Inverter



A	Out
F	T
T	F



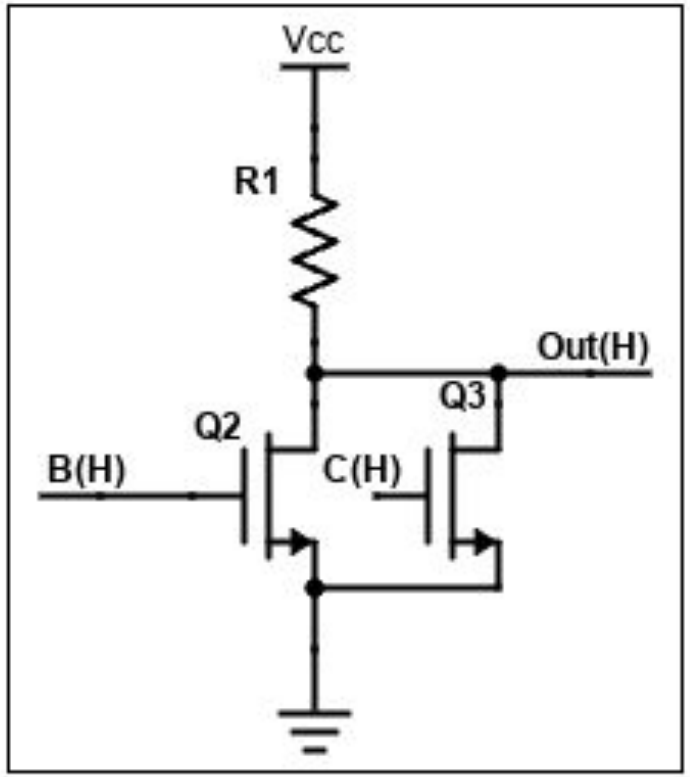
NMOS NAND Gate



Input 1	Input 2	NAND
F	F	T
F	T	T
T	F	T
T	T	F

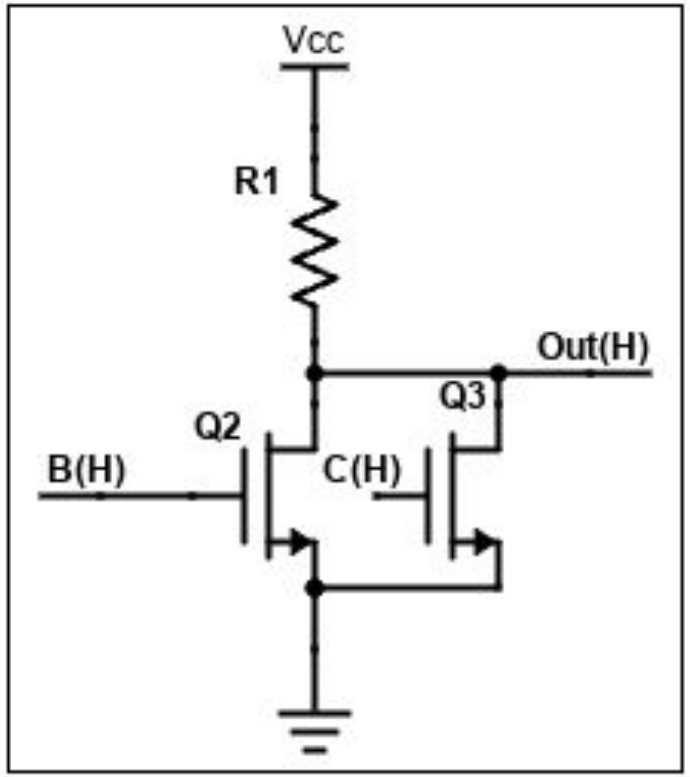


Who's that Logic Gate?

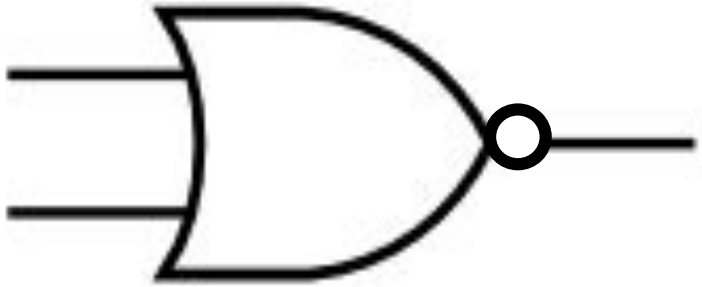


B	C	Out
F	F	?
F	T	?
T	F	?
T	T	?

NMOS NOR Gate

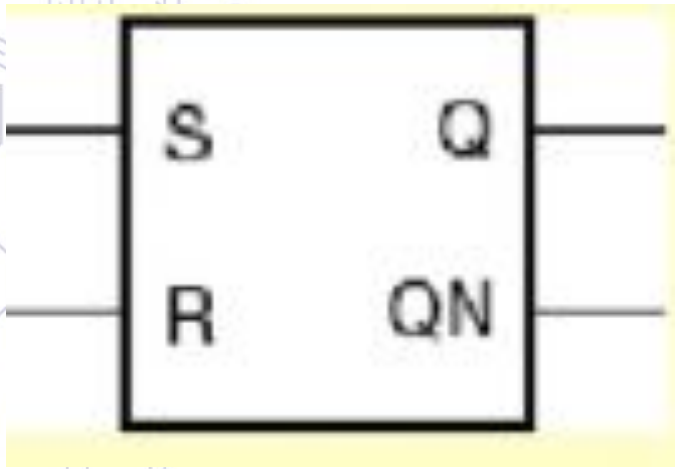


B	C	Out
F	F	?
F	T	?
T	F	?
T	T	?



Project 4: SR Latch

SR Latch



- A Set-Reset Latch (SR Latch) is a 1-bit asynchronous memory
- The SR Latch is a fundamental building block of digital circuits

SR Latch Truth Table

S	R	Q	Q’	State
F	F	Q	Q’	Hold
F	T	0	1	Reset
T	F	1	0	Set
T	T	X	X	Invalid

- S: Set the memory to ‘1’
- R: Reset the memory to ‘0’
- When neither S nor R are true, the latch holds it’s stored value
- When both S and R are true, the output is undefined

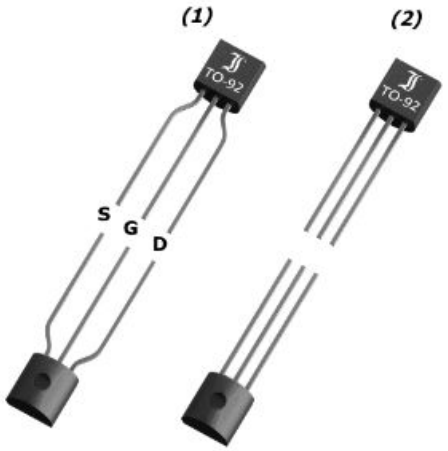
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NMOS Pinout

2N7000
N-Channel Enhancement Mode FET
N-Kanal FET – Anreicherungstyp

Version 2025-01-31

TO-92 10D3



Typical Applications
Signal processing
Logic level converter
Drivers
Commercial grade
Suffix -Q: AEC-Q101 compl
Suffix -AQ: in AEC-Q101 qu

Features
Fast switching times
Compliant to RoHS (w/o ex
REACH, Conflict Minerals [1])

Mechanical Data [1])

(1) Taped in ammo pack

- **Your OPS kit contains 4 2N7000 NMOS Transistors**
- **Check the pinout from the datasheet**
- **Pins are labelled based on the flat side facing you**

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Keep up with IEEE!

MON	TUE	WED	THU	FRI	SAT
2	3 7pm Arts and Crafts Ceramic Soc	4 10:45am Curriculum Office Hours	5	6	7
9 10am Mill Creek Horse Farm	10	11	12 6pm GBM #4	13	14 1pm IEEE x WECE Thanksgiving P
16	17	18 10:45am Curriculum Office Hours	19 6pm IEEE Seminar: Ariel David Ce	20	21
23	24	25	26	27	28
30	Dec 1	2 10:45am Curriculum Office Hours	3	4 6pm IEEE vs Faculty Volleyball	5

Upcoming events:

11/12: GBM #4

11/14: Thanksgiving Potluck

11/19: IEEE Seminar

Reminders:

HAVE A HAPPY THANKSGIVING!!!

Project 4 due 12/1/26!

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